WORKED SOLUTION

# STEP II 2023, Question 10

Mechanics • Projectile Envelope

**STEP II 2023, Question 10**

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

## The key idea

A trajectory and the curve $E$ *touch* exactly when the equation for their common points has a repeated root, so part (i) is a discriminant calculation. Asking for this to happen for every angle forces $E$ to be the *envelope* of all the trajectories: the boundary of where particles of speed $u$ can reach. The rest of the question looks at that boundary from three angles: which points can be hit, where the particles are at a fixed time, and a second particle sliding along the envelope itself.

## Solution

Throughout, the particles are modelled as points moving freely under gravity, with no air resistance, and $c = \cos\alpha$, $s = \sin\alpha$.

### Part (i)

With $t$ the time since projection, $P_\alpha$ stays in the plane $y = 0$ with

$$x = uct, \qquad z = ust - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(uc)$ gives the trajectory

$$z = x\tan\alpha - \frac{gx^2}{2u^2c^2}.$$

Common points with $E$ satisfy $A - Bx^2 = x\tan\alpha - \dfrac{gx^2}{2u^2c^2}$, that is

$$\left(\frac{g}{2u^2c^2} - B\right)x^2 - x\tan\alpha + A = 0. \qquad (*)$$

If the coefficient of $x^2$ were zero, $(*)$ would be linear with $\tan\alpha \neq 0$, giving a single crossing rather than a touching, so $(*)$ is a genuine quadratic. The curves touch exactly

once when $(*)$ has a repeated root, i.e. when its discriminant vanishes:

$$\tan^2\alpha = 4A\left(\frac{g}{2u^2c^2} - B\right).$$

Multiplying by $u^2c^2$:

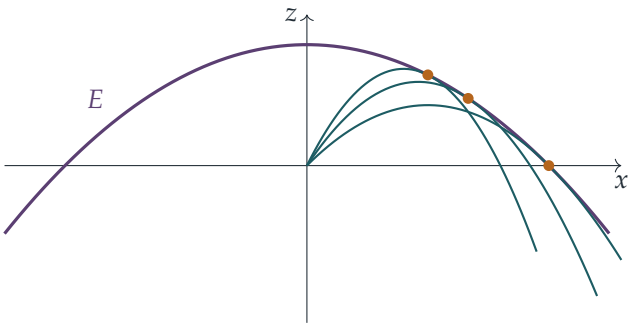
$$\begin{aligned} u^2s^2 &= 2gA - 4ABu^2c^2 \\ u^2 - u^2c^2 &= 2gA - 4ABu^2c^2 \qquad (\text{using } s^2 = 1 - c^2) \\ u^2 - 2gA &= u^2(1 - 4AB)\cos^2\alpha, \end{aligned}$$

as required.

If this holds for every $\alpha$ in $(0, \frac{1}{2}\pi)$, then the left-hand side is a constant while $\cos^2\alpha$ takes every value in $(0, 1)$. A constant equal to $k\cos^2\alpha$ for all these values forces $k = 0$ and then the constant is 0. So $u^2 = 2gA$ and $4AB = 1$:

$$A = \frac{u^2}{2g}, \qquad B = \frac{g}{2u^2}.$$

Conversely these values make the identity true for every $\alpha$. The touching point is the repeated root $x = \dfrac{\tan\alpha}{2(g/(2u^2c^2) - B)} = \dfrac{u^2}{g\tan\alpha}$.




The figure (drawn with $u^2/g = 2$) shows three trajectories each touching $E$ once, at the amber points.

**Part (ii)**

Every particle moves in the vertical plane containing its initial velocity, and everything is symmetric about the $z$-axis. So take a point at horizontal distance $r$ from the $z$-axis and height $z$, and work in the vertical plane through it, with $\theta \in (-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the angle of projection above the horizontal towards the point.

If $r = 0$, only particles fired straight up or down can arrive; the one fired upwards reaches every height up to $u^2/(2g)$ and the one fired downwards every negative height, so the reachable points on the axis are those with $z \le u^2/(2g)$.

If $r > 0$, exactly as in part (i) the particle passes through the point if

$$z = r\tan\theta - \frac{gr^2}{2u^2}(1 + \tan^2\theta),$$

using $1/\cos^2\theta = 1 + \tan^2\theta$. Writing $T = \tan\theta$, which can take any real value, this is the quadratic

$$\frac{gr^2}{2u^2}T^2 - rT + \left(z + \frac{gr^2}{2u^2}\right) = 0,$$

which has a real root precisely when

$$r^2 \geq \frac{2gr^2}{u^2}\left(z + \frac{gr^2}{2u^2}\right) \iff z \leq \frac{u^2}{2g} - \frac{gr^2}{2u^2},$$

dividing by $2gr^2/u^2 > 0$.

So, with $r^2 = x^2 + y^2$, the points that can be hit are exactly those on or below the paraboloid

$$z = \frac{u^2}{2g} - \frac{g}{2u^2}(x^2 + y^2),$$

which is the surface obtained by rotating $E$ (with the values of part (i)) about the $z$-axis. Points strictly below it are reached by two directions in the plane, points on it by exactly one (the repeated root, which matches the touching in part (i)), and points above it by none.

#### Part (iii)

Let the particle have initial velocity $u\mathbf{n}$, where $\mathbf{n}$ is a unit vector. After time $t$ its position is

$$\mathbf{r} = ut\,\mathbf{n} - \tfrac{1}{2}gt^2\,\mathbf{k},$$

where $\mathbf{k}$ is the upward unit vector. Hence

$$|\mathbf{r} - (-\tfrac{1}{2}gt^2\,\mathbf{k})| = |ut\,\mathbf{n}| = ut.$$

So at time $t$ every particle lies on the sphere with centre $(0, 0, -\frac{1}{2}gt^2)$ and radius $ut$. Physically: gravity accelerates every particle equally, so relative to a point falling freely from the origin they all fly outwards in straight lines at speed $u$.

#### Part (iv)

With $\tau$ the time since $Q$ was projected, $Q$ has $y = 0$ and

$$x = u\tau, \qquad z = A - \tfrac{1}{2}g\tau^2.$$

Substituting $\tau = x/u$:

$$z = A - \frac{g}{2u^2}x^2 = A - Bx^2,$$

using $B = g/(2u^2)$ from part (i). So $Q$ is always on $E$.

**Part (v)**

Suppose $Q$ is projected at time $T$ after the explosion, and that $Q$ and $P_\alpha$ are at the same point at time $t$ ($\geq T$) after the explosion. Then $\tau = t - T$ and both $y$-coordinates are 0, so we need the $x$- and $z$-coordinates to agree:

$$uct = u(t - T),$$
$$ust - \tfrac{1}{2}gt^2 = A - \tfrac{1}{2}g(t - T)^2.$$

The first gives $t - T = ct$, so $T = t(1 - c)$. Putting $t - T = ct$ into the second and using $1 - c^2 = s^2$:

$$ust - \tfrac{1}{2}gs^2t^2 = A = \frac{u^2}{2g}.$$

**Ask yourself** How do we solve this for $t$ without a messy formula?

Multiply by $2g$ and collect everything on one side: $g^2s^2t^2 - 2gus\,t + u^2 = 0$, which is the perfect square

$$(gst - u)^2 = 0, \qquad \text{so} \qquad t = \frac{u}{g\sin\alpha}.$$

A repeated root is just what we should expect: the collision has to happen where the trajectory of $P_\alpha$ touches $E$, since that is the only point the two paths share. Indeed $P_\alpha$ is then at $x = uct = u^2/(g\tan\alpha)$, the touching point found in part (i). Therefore

$$T = t(1 - \cos\alpha) = \frac{u(1 - \cos\alpha)}{g\sin\alpha},$$

as required. Since $0 < 1 - \cos\alpha < 1$, this gives $0 < T < t$, so $Q$ is indeed launched after the explosion and before the collision.

## Checks

- In part (i), $\alpha = \frac{1}{4}\pi$ gives the touching point $x = u^2/g$, $z = 0$: the maximum range on level ground, which the boundary of the reachable region should indeed pass through.
- In part (ii), on the axis $r = 0$ the bound is $z \leq u^2/(2g)$, the greatest height of a particle fired straight up.
- In part (v), with $u = 7$, $g = 9.8$ and $\alpha = 0.5$, the formula gives $T \approx 0.182$; stepping both motions forward in time confirms that the particles meet at $t = u/(g\sin\alpha) \approx 1.49$, and that launching $Q$ even 0.05 later makes them miss.
- As $\alpha \to \frac{1}{2}\pi$, $T \to u/g$ and $t \to u/g$: $P_\alpha$ goes almost straight up to the top of $E$ at $(0, 0, A)$, arriving at time $u/g$, just as $Q$ sets off from there.

## Common slips

- Setting the discriminant to zero without first rewriting the common-points equation as a single quadratic in $x$, and so dropping the $-B$ term from the $x^2$ coefficient.
- In part (ii), describing only the boundary surface: the question asks for the whole set, which is the solid region on and below the paraboloid.
- In part (v), using the same time variable for both particles: $Q$ has only been moving for $t - T$.

## Beyond STEP

The curve $E$ is the *envelope* of the family of trajectories: a curve tangent to every member of the family. In general the envelope of a family $F(x, z, \alpha) = 0$ is found by eliminating $\alpha$ between $F = 0$ and $\partial F/\partial \alpha = 0$, which is the calculus version of the repeated-root condition used here. The paraboloid in part (ii) is often called the *paraboloid of safety*, and part (iii) is an example of changing to a freely falling frame, where gravity disappears. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas.

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